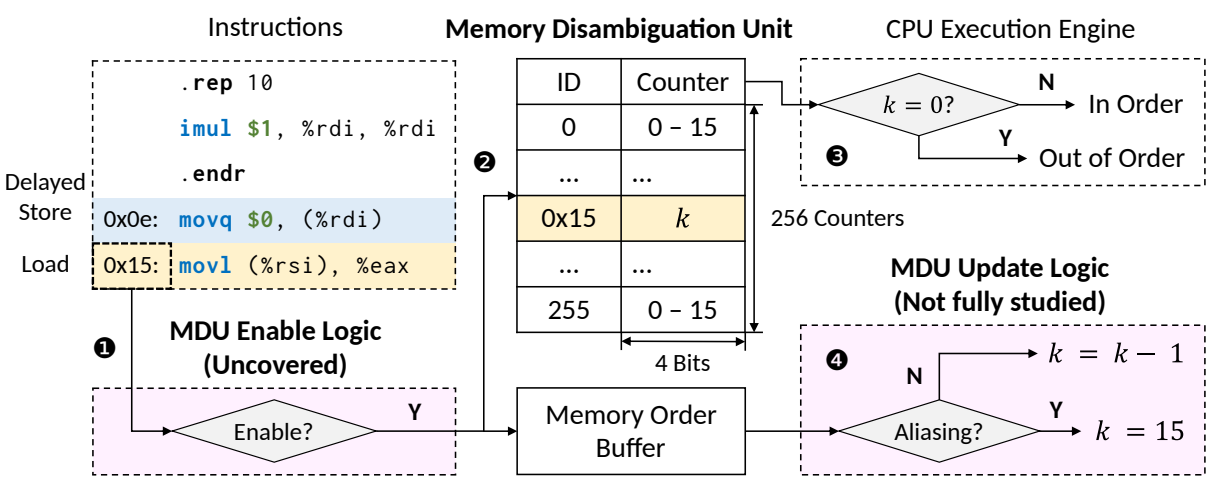
Instructions
Memory Disambiguation Unit
CPU Execution Engine
.rep 10
imul $1, %rdi, %rdi
.endr
0x0e: movq $0, (%rdi)
0x15: movl (%rsi), %eax
Delayed Store
Load
ID
Counter
0
0 – 15
...
...
0x15
k
...
...
255
0 – 15
256 Counters
4 Bits
k = 0?
N
Y
In Order
Out of Order
❶
MDU Enable Logic
(Uncovered)
Enable?
Y
❷
Memory Order Buffer
❸
❹
MDU Update Logic
(Not fully studied)
N
k = k − 1
Aliasing?
Y
k = 15